

Nicolás Aller Ponte

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Education

Universidade da Coruña, in Bachelor's Degree in Data Science and Engineering – A Coruña, Spain Sept 2022 – present

- Knowledge of the complete data lifecycle: HPC, statistics, data engineering, databases and AI.

Experience

Research Assistant, Inditex-UDC Chair of AI in Green Algorithms – A Coruña, Spain Nov 2025 – Mar 2026

- Research and development of efficient algorithms for AI models.

AI Engineer, Gradiant – A Coruña, Spain Sept 2025 – Oct 2025

- Development of a car damage detection system.
- Development of an evaluation system for car accident reports.

Languages

Proficiency: Spanish (Native), Galician (Native), English (Intermediate), German (Basic)

Skills

Data Engineering & Cloud: Python, R, SQL, MongoDB, Docker, Git, Spark, AWS, GCP

AI & Deep Learning: PyTorch, TensorFlow, Keras, Scikit-learn, Computer Vision, NLP

HPC & Systems Programming: C, C++ , Parallel Computing (OpenMP, MPI), Computer Architecture, CUDA

Awards

First Place — Best Use of Open Source AI (HackUDC 2026): Recognition for the integration and optimization of open-source models (SigLIP, DINOv2) in a scalable production environment

Second Place — Inditex-Tech Challenge (HackUDC 2026): Finalist for developing a visual search engine for massive fashion catalogs

Second Place — GDG Challenge (Impacthon 2026): Runner-up for developing LocalFold, a web portal for protein structure prediction using AlphaFold2, A2A multi-agent system, and Gemini-powered analysis on Google Cloud

Projects

Outfit Intelligence — HackUDC – A Coruña, Spain Mar 2026

- Multi-layer visual search pipeline using YOLOv8 and SigLIP-384 to identify catalog products in outfit images.
- Model fine-tuning with contrastive loss and cross-validation, combining SigLIP and DINOv2 in an ensemble to improve accuracy.
- Recommendation engine using FAISS and a REST API with FastAPI on a catalog of 27,000+ products.

LocalFold — Impacthon 2026 – GDG Challenge Apr 2026

- Web portal for protein structure prediction via AlphaFold2 without HPC expertise, featuring interactive 3D visualization (Mol*), PAE heatmaps, and pLDDT confidence coloring.
- A2A multi-agent system with four specialist agents (mutation, lab conditions, clinical relevance, experimental design) powered by Gemini 2.5 Flash with Google Search Grounding.
- Deployed on Google Cloud Run with FastAPI, Firestore, and six-language UI support.